

Experiment No. 4

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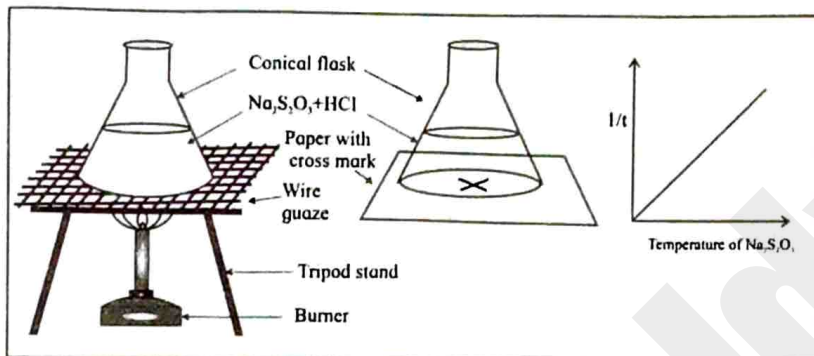
Aim: To study the effect of temperature on the rate of reaction between sodium thiosulphate and hydrochloric acid.

Theory: The rate of a chemical reaction depends upon temperature. As the temperature increases the rate of reaction increases due to increase in kinetic energy of the molecules. The increase in kinetic energy of molecules increases the number of effective collisions between reacting species.

Apparatus: Burettes (50 mL), conical flask(250 mL), stop watch, plane paper with cross marks, thermometer (360°C), wire gauze, tripod stand, measuring cylinder (10 mL), etc.

Chemicals: 0.1 M $\text{Na}_2\text{S}_2\text{O}_3$, 1 M HCl, distilled water

Diagram:



Procedure:

1. Clean the conical flask and take 50 mL 0.1 M $\text{Na}_2\text{S}_2\text{O}_3$ solution. Record its initial temperature using thermometer (t °C). Keep the conical flask on paper with cross mark.
2. Add 2 mL of 1M HCl to the conical flask, and start stop watch immediately shake well and record the time required for the colloidal sulphur formed during the reaction to hinder the vision of cross mark.
3. In another conical flask take 50 mL 0.1 M $\text{Na}_2\text{S}_2\text{O}_3$ solution and heat it to increase the temperature of solution by 10 °C [$(t + 10)$ °C].
4. Remove the flask and keep it on paper with cross mark and add 2 mL of 1 M HCl solution, shake well, start the stopwatch. Record the time required when cross mark on the paper just become invisible.
5. Repeat the procedure at $(t+20)$, $(t+30)$ and $(t+40)$ °C and record the time as before.

Observation Table:

Volume of 0.1 M $\text{Na}_2\text{S}_2\text{O}_3$ 50 mL

Volume of 1 M HCl added 2 mL

Temperature °C	Time required when cross mark just becomes invisible in seconds	$1/t \text{ s}^{-1}$
$t = 25$	74	0.015
30 (t)+10 =	44	0.023
40 (t)+20 =	33	0.03
50 (t)+30 =	27	0.037
60 (t)+40 =	22	0.045

Graph: Plot a graph of $1/t$ against the temperature of $\text{Na}_2\text{S}_2\text{O}_3$ solution.

Result: Rate of reaction between $\text{Na}_2\text{S}_2\text{O}_3$ and HCl ... increase with increase in temp.

Remark and sign of teacher:

MCQ

Select the correct answer for the following multiple choice type questions.

1. What happens when we increase the temperature of the reaction mixture of aqueous sodium thiosulphate and hydrochloric acid solutions?
 - a. it forms clear solution
 - ☒ b. collision between reacting species increases
 - c. kinetic energy of reacting species decreases
 - d. it forms black coloured solution
2. The nature of slope of the graph in above experiment?
 - a. decreases
 - ☒ b. increases
 - c. first decreases and then increases
 - d. first increases and then decreases
3. Which one of the following factor on which the rate of reaction between 0.1M $\text{Na}_2\text{S}_2\text{O}_3$ and 1M HCl solutions in above experiment depends?
 - a. catalyst
 - b. pressure
 - c. concentration
 - ☒ d. temperature

Short answer questions

1. How would you do the experiment if there is no thermostat available in your laboratory?

Ans. If no thermostat system is available, there are two simpler, but not as accurate alternatives. If the experiment doesn't take too long, a large beaker or trough of water might be used to cool the reaction mixture.

2. As the temperature increases, the rate of reaction increases. Explain why?

Ans. As the temperature increases, the rate of reaction increases due to an increase in the kinetic energy of the molecules.

3. What will happen, when kinetic energy of the reacting molecules increases?

Ans. The increase in kinetic energy of molecules increases the number of effective collisions between the reacting molecules.

4. How does the rate constant of reaction vary with temperature?

Ans. The rate constant of a reaction varies with temperature.

5. Why does sodium thiosulphate give colloidal precipitate on reaction with HCl?

Ans. Sulphur is formed during the reaction, which is insoluble and forms a milky non-transparent aqueous colloidal solution.

Remark and sign of teacher: